A Town Gate That Learns to Wait

A plain-language account of the "stigmergic-quorum admission control" technique

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The problem. Some disputes that reach a moderation system are genuinely hard to settle: two sides each have arguments that attack the other's, and the easy, fast method for weighing them comes back with no clear answer. When that happens, the system falls back to a much more thorough, much more expensive method of sorting the whole tangle out — worth doing, but not something you want to do for free, on demand, for anyone who asks. The trouble is in how that expensive method gets triggered. In the original design, the very first person who shows up with a plausible-looking argument gets the case waved through to the expensive process, instantly. That is exactly the kind of rule an attacker loves: show up once, in bulk, with a pile of manufactured arguments, and the expensive gate opens for free, as many times as you like.

The idea. The fix borrows two tricks from nature, and it is worth taking them one at a time, because each solves a different half of the problem.

The first trick is how ant colonies remember which path to food is worth following. An ant does not vote once and settle the matter; it lays down a scent as it walks, and that scent fades steadily unless other ants, arriving independently, walk the same way and refresh it. A path only stays marked if it keeps earning fresh footsteps from different ants — one ant pacing back and forth on its own trail all day leaves a trail that is, for practical purposes, exactly as strong as one visit. Applied here: every disputed case carries a score that fades a little every tick of the clock, and a new argument only tops the score back up by very little if it comes from a source that has already weighed in recently. Genuinely new, independent concern keeps a case's score alive; the same complainer shouting the same thing over and over does not.

The second trick is how bacteria decide, as a colony, when there are enough of them to act together — a mechanism called quorum sensing. No single bacterium makes that call alone; the whole colony reads a shared chemical signal that reflects how crowded things already are, and raises or lowers its collective threshold for action accordingly. Applied here: the bar a case must clear to reach the expensive process is not fixed. It rises automatically whenever that expensive process is already busy with other cases, exactly the way a restaurant becomes harder to get a table at as it fills up, without anyone at the door consciously deciding to be stricter.

Put together. A disputed case's score decays like an ant trail losing its scent, is topped up only by reinforcement that looks like it came from someone new, is capped so no single moment of activity can push it past a sensible cap, and must clear a bar that rises with how busy the expensive process already is. A single burst of manufactured outrage, however large, cannot move a decaying, capped score past a threshold in one instant — there simply is not enough weight allowed into any one tick of the clock, no matter how many fake voices show up at once.

What was actually measured, not just hoped for. The mechanism was tested against its own predecessor — the instant-admission rule — under several kinds of attack, and the two were compared honestly, including on the case where the new mechanism does *not* fully win. A single flood, of any size from one voice to five thousand, landing in one moment: fully blocked, every time, where the old rule let every single one through. The same flood

sustained for a very long time from one traceable source: still fully blocked, provably, not just "blocked so far in the tests that were run." Two hundred separate disputes each hit with a one-shot burst, alongside fifteen genuine, organically-growing disputes running at the same time: the old rule let every attack through and admitted every genuine case too, at a real processing cost of 215 cases; the new mechanism let through zero of the attacks while still admitting every one of the genuine fifteen, at a real processing cost of 15 — ninety-three percent less work forced onto the expensive process. The honest exception, stated plainly rather than buried: a patient attacker willing to manufacture a fresh, never-reused, independent-looking identity every single tick of the clock, for several ticks in a row, is not fully stopped. That attacker still gets through eventually, at roughly five times the effort and time the old rule would have cost them. The mechanism turns a free attack into an expensive one; it does not turn it into an impossible one, and that limit is reported here rather than left for someone else to discover the hard way.

What it deliberately does not do. It says nothing about who is *right* in the underlying dispute — that judgment is left entirely to the expensive process it is guarding, once a case is let through. It assumes there is some real cost, even a small one, to appearing as a brand-new independent voice every single tick; if that assumption ever stops holding for a given attacker, the delay is the whole defense, not a backup behind something stronger. And the fading-and-capping trick and the rising-bar trick, taken completely apart from each other, are each individually well known outside biology too — the first resembles an ordinary rate limiter used all over computing, the second resembles a decades-old networking technique for easing congestion before a queue overflows. What a dedicated search of the relevant research literature did not turn up is this particular combination of all three pieces, guarding specifically this kind of expensive, hard-to-parallelize fallback process. That is the narrow claim being made, and only to the extent that search was thorough — not a claim that no one, anywhere, has ever combined these ideas before.

Why it is worth writing down. Nothing about the expensive process itself needed to change; the fix sits entirely at the door in front of it. The individual ingredients — a fading score, a cap on how much one moment can contribute, a bar that rises with crowding — are each borrowed from places with decades of their own track record, biological and computational alike. What is reported here is how they compose, tested honestly against the attack it was built to resist, with the case where it only partly succeeds stated as plainly as the cases where it fully does.